Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. TEAM AND PROJECT OVERVIEW

Team Name

SDC4

Selected Domain

Domain 1

Selected SDG Track (1–17)

4 Quality education, 8 Decent Work and Economic Growth

Project / Solution Name

NEXTEP

Name (Leader)

YUE XIONGHE

Email & Phone (Leader)

ericyywies@qq.com 1123410321

University Affiliation(s)

XIAMEN UNIVERSITY

Number of Team Members

4

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1. The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

As Electrical and Electronic Engineering (EEE) students, we witness firsthand how rapid AI advancements create severe career turbulence for engineering students. Traditional academic curricula lag behind industry demands, leaving students uncertain about which future skills to pursue amidst workforce disruption.

We chose EEE because it is not only our academic foundation, but also the essential technological bedrock powering solutions across all Sustainable Development Goals (SDGs). EEE innovations directly drive sustainability—from ultrasonic navigation tools that reduce inequalities (SDG 10) and air-quality sensors for smart cities (SDG 11), to flood-warning electronics (SDG 13), ocean buoys (SDG 14), and open engineering datasets (SDG 17).

Without transparent, data-driven career navigation, engineering students face structural skill mismatches, undermining both their employability and this broader societal impact. Empowering students with actionable reskilling pathways directly bridges this gap. By guiding the next generation toward emerging EEE domains like green tech, AI hardware, and edge sensing, we advance quality education (SDG 4), foster productive employment (SDG 8), and equip future engineers to build a sustainable world.

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.2. The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

To address the growing career friction among Electrical and Electronic Engineering (EEE) students, we present NEXTEP—an AI-driven career navigation platform designed to bridge the gap between academic education and industry demand. NEXTEP analyzes massive real-time recruitment datasets and market trends to uncover hidden career opportunities and pivotal transition paths for individuals amidst rapid industry shifts.

Our application leverages Generative AI APIs (powered by DeepSeek) to evaluate individual skill profiles, match them against emerging engineering roles, and automatically generate personalized, step-by-step learning roadmaps. This empowers students to systematically close their skill gaps and achieve their long-term career goals.

In alignment with the technical skills gained during the SDC Phase 1 workshops, NEXTEP incorporates a full-stack architecture:

GenAI API Integration: Utilizing LLMs for automated skill extraction, intelligent profile matching, and customized career trajectory advice.

Database Architecture & Big Data Analytics: Structuring open-source recruitment datasets to process market signals and power accurate semantic skill mapping.

Full-Stack Development & Cloud Hosting: Built with Next.js for a responsive UI and deployed on Replit for seamless cloud testing and high availability.

By combining big data analytics with GenAI intelligence, NEXTEP turns career volatility into personalized opportunity—guiding EEE students toward a resilient future.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

NEXTEP is a web prototype built with Next.js 16, React 19 and TypeScript, and hosted on Replit. Its API layer serves country-specific job-market datasets covering China, Europe, Malaysia and the United States. The prototype uses rule-based methods for EEE role matching, skill-gap analysis and personalised learning-path generation, alongside team-curated mappings to relevant UN Sustainable Development Goals. Student profiles and learning progress are stored in the browser using localStorage and IndexedDB; a central user database is not required for the current Replit deployment. An optional server-side DeepSeek integration is available but disabled by default and is not required by the core features. Career outlooks are scenario-based results, not hiring probabilities or validated forecasts.

Yes. We used Replit as the cloud development and hosting platform for the NEXTEP web prototype. Our team received a US$20 Replit credit voucher from the SDC sponsor. We did not use an SDC-provided GenAI API.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.2. Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

To ensure NEXTEP is feasible, intuitive, and ready for Phase 2 deployment, our team applied a 5-stage Design-Thinking methodology:

1. Empathize & Define: As EEE students, we conducted peer interviews to map the anxiety and friction caused by AI disruption, pinpointing the core need: transparent, actionable skill transition pathways.

2. Ideate: We brainstormed AI-driven solutions, deciding to combine LLM intelligence (DeepSeek API) with real-time recruitment datasets to transform career market signals into personal guidance.

3. Prototype: We built an interactive full-stack web application using Next.js and Tailwind CSS, focusing on a clean, accessible UI that converts complex skill analysis into step-by-step roadmaps.

4. Test & Deploy: Using Replit for continuous integration and cloud hosting, we iteratively tested API responses and user workflows, validating technical viability and real-time performance.

For Phase 2, our deployment strategy focuses on integrating live API data pipelines, enhancing personalized feedback loops, and gathering real student user testing to refine the roadmap recommendation engine.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 4.4: "By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship." Target 8.6: "By 2030, substantially reduce the proportion of youth not in employment, education or training (NEET)."
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	According to global education and industry surveys, over 65% of graduating engineering students report severe career anxiety and uncertainty due to AI disruption, while 42% of EEE graduates experience a structural skill mismatch between university curricula and emerging high-tech industry demands during their first job search.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided*	Primary KPI: % improvement in skill-to-job profile match accuracy for EEE students. Secondary KPIs: % increase in student confidence/clarity regarding tech transition paths, and average time saved (in hours/ week) in identifying targeted upskilling resources.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	30% improvement in skill match accuracy between student profiles and emerging EEE job market requirements, along with a 25% reduction in time spent by students searching for relevant reskilling resources during the final prototype testing.

5. ELIGIBILITY & READINESS CHECKLIST

Tick all boxes that apply before submitting.

☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP & ORIGINALITY DECLARATION

6.1. Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6. IP & ORIGINALITY DECLARATION

6.2. Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Next.js, React, React DOM, Vite	Frameworks and build tools	MIT
Base UI, Radix UI, shadcn/react, Recharts, Lucide React, and other UI libraries	UI, charting and icon libraries	Mostly MIT; Lucide React: ISC. See each package's license.
TypeScript; class-variance-authority; Drizzle ORM; PDF.js	Developer tools and libraries	Apache-2.0
United Nations SDG goal definitions	Official reference content	Cite the relevant UN goal pages; NEXTEP's role-to-SDG mappings are team-authored and are not endorsed by the UN.
Mammoth; PptxGenJS	Document import/export libraries	Mammoth: BSD-2-Clause; PptxGenJS: MIT

6. IP & ORIGINALITY DECLARATION

6.3. Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

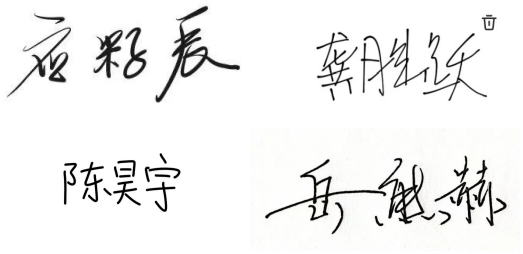
☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]



陈昊宇

[Date]

24/10/2026